CURRICULUM VITAE

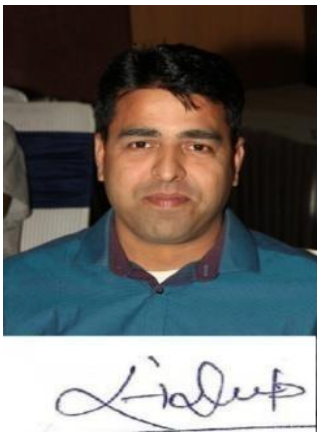
DR. SANDEEP KUMAR OJHA

Assistant Professor (Ad hoc)

Department of Electronics and Communication Engineering,

Visvesvaraya National Institute of Technology, Nagpur

Nagpur-440010 (Maharashtra) India



CONTACT DETAILS

Contact No.- **+91-9140536146**

Address: Sector-2, Shivaji Nagar, P.O.-Barago, Ranibagh, Rustampur, Gorakhpur- 273016, INDIA

E-mail: sandeepojhasandeep@gmail.com

AREA OF RESEARCH INTEREST:

Internet of Things (IoT), Arduino Uno, Raspberry Pi, Microprocessor and Microcontroller, VLSI and Microelectronics: VLSI Design and Technology, Electronics Devices and Circuits, Digital Circuits, Semiconductor Electronics.

Organic Electronics: Organic light emitting diode (OLED), Organic light emitting transistor (OLET), Organic solar cells (OPV), Modelling and Simulation of electronics devices.

SCHOLASTICS

- **Ph.D.** Completed in the area of Microelectronics & VLSI under supervision of Prof. Brijesh Kumar, Professor, ECE Department at M.M.M.U.T., Gorakhpur,

  Thesis Title: Analytical Modelling and Parameter Extraction of Organic Light Emitting Transistor (OLET).

- **M.Tech. with HONOURS** in (VLSI Design) from SIIT, Allahabad, Affiliated by UPTU, Lucknow in 2015 secured 75.05 %.

  M.Tech. Specialization: Design, Test, Testability and Fault Detection in Analogue and Digital VLSI Circuits.

- **B.Tech. (Electronics & Communication)** from J. K. Institute of Applied Physics & Technology, University of Allahabad in 2001 Secured 63%.

- **B.Tech. Specialization**: Study of digital modulation (DM) & design of pulse code modulation (PCM) communication-system.

  Types of modulation, various digital communication system & schemes, theory of PCM, circuit description & IC specification.

- **B.Sc.** with Mathematics and Chemistry from St. Andrew's College, University of Gorakhpur in 1998. Secured 75% (**2nd Topper in whole Gorakhpur Zone**).
- **Intermediate** from U.P. Board, Allahabad, Uttar Pradesh, INDIA with Second Division.
- **High School** from U.P. Board, Allahabad, Uttar Pradesh, INDIA with First Division.

CAREER CONTOUR

- **Assistant Professor (Ad hoc)** at VNIT, Nagpur Since 25th July 2025.
- **Assistant Professor (Guest Faculty)** at MMMUT, Gorakhpur from 1st Sept. 2022 to 18th Jan 2025.
- **Assistant Professor** at ITM, GIDA, Gorakhpur since May 2018 to Nov. 2021. Teaching to students Basic Electrical Engineering, Digital communication, Signal System and done counseling for Admission to all students related to different courses.
- **Associate Professor** at KIPM Technical Campus, GIDA, Gorakhpur from Aug. 2016 to May 2018. Counseling for admission in different courses.
- **Asst. Professor** at SAITM, Farukhnagar, Gurgaon from August 2014 to Aug. 2016. Teaching to students Basic Electronics and Electrical Technology, Digital & Analog Communication, Principles of Communication System.
- **Asst. Professor** at LDCITS, Soraon, Allahabad, U.P. since July 2011 to August2014, Teaching to students Basic Electronics and FED.
- **Asst. Professor** at GITM, Gurgaon Haryana July 2010 -July 2011-Teaching to Students Basics Electronics and Control System.
- **Manager (O&M)** at Reliance Communication Limited June 2009-July 2010 for Azamgarh, Mau and Balia districts in U.P (600 sites).
- **Sr. Executive** (Operation & Maintenance) at Indus Tower Limited. Pune, Maharashtra, Jan 09-June 09 for 7 District in Maharashtra (1000 Sites).
- **Circle Head** (Zonal Head O&M) at Emerson Network Power ENP (India) Pvt. Ltd.: Jan 2007-Jan 2009- O & M functioning of D.G., SMPS, A.C., PIU, UPS, IT Servo, PIU & Shelter.
- **Projects Handled**: Any academic projects like Industrial visit, counseling and administration of students completed by me. During TTSL Project in Delhi / NCR Duration: Jan'07- Dec'07. Vodafone Project in U.P. East Gorakhpur Zone Duration: Dec. 2007-25Th Jan. 2009 (handling 1000 sites over 7 districts) **Received Best Cluster Head Award in 2006-2007 from Dubai ENP.**
- **High Tension (HT) Project**: Handling Assembling & Installation of 11 KVA HT Panel approved by ABB (Asia Brown Baweri Ltd. Nasik.

CAREER & ACADEMIC COUNSELLOR

As a Career & Academic Counsellor, I guide students in making informed educational and career decisions aligned with their interests, aptitude, and industry requirements. I mentor Intermediate and undergraduate students on higher education opportunities, research pathways, competitive examinations, and career planning.

ACHIEVEMENTS & AWARDS

- **Best Cluster Engineer Award-2007 from DUBAI EMERSON GROUP** .
- During college time Achieve **1st prize in Solo song, Duet song, Group song and 2nd prize in Skit and Choreography.**

LIST OF JOURNAL AND CONFERENCE PUBLICATION

INTERNATIONAL JOURNAL PUBLICATIONS (SCI)

[1] **Sandeep Kumar Ojha** and Brijesh Kumar, "Parameter Extraction of High-Performance Material Based Organic Light-Emitting Transistors (OLETs)", **Silicon (ISSN: 1876-9918)**, Vol. 49, 2021, https://doi.org/10.1007/s12633-021-01149-6 **(Published by Springer, SCI, Peer reviewed, Impact Factor: 4.1)** https://doi.org/10.1007/s12633-021-01149-6.

[2] **Sandeep Kumar Ojha** and Brijesh Kumar, "Coulomb Blockade Effect through Single Electron Tunneling Method in Cylindrical Gate Organic Light Emitting Transistor Configuration", **Silicon (ISSN: 1876-9918)**, Vol. 42, 2021, https://doi.org/10.1007/s12633-021-01190-5 (**Published by Springer, SCI, Peer reviewed, Impact Factor: 4.1)**.

[3] **Sandeep Kumar Ojha** and Brijesh Kumar, "Analysis of Electrical Characteristics and Electroluminescent Efficiency of Field Induced Contact-DGOLET", **Main Group Chemistry** (ISSN:1745-1167), pp. 1-11, January 1, 2022, (**Published by IOSPress, SCI, Indexed by Web of Science, Scopus, Peer reviewed, Impact Factor: 1.8), Doi**: 10.3233/MGC-210144.

[4] Dinesh Chandra, Anupam Shahu, Brijesh Kumar, Aryan Kumar Kannaujiya and **Sandeep Kumar Ojha**, "Performance Enhancement of OLED Employing CuSCN Interfacial layer" **Journal of Optics (ISSN 0974-6900), (Published by Optics Jornal, SCI, Peer reviewed, Impact Factor: (2.6) ID: OPTI-D-23-01173, 29/12/2023**.

[5] Shekhar Yadav, Jai Prakash Pandey, Rajesh Kumar Yadav, **Sandeep Kumar Ojha,** Brijesh Kumar, "Method to determine the maximum current density of p-

MoS2/ p-CdSe/n-WS2/n-ITO solar cell" **Journal of Optics (ISSN 0974-6900),** (**Published by Optics Jornal, SCI, Peer reviewed, Impact Factor: (2.6)** https://doi.org/10.1007/s12596-024-02291-1, **27 September 2024.**

[6] Rajesh Kumar Tiwari[1], Rajesh Mishra[2], Sanjay Kumar Sharma[3], **Sandeep Kumar Ojha[4]**, Anushka Ojha[5], "Enhancing Prostate-Specific Antigen (PSA) Density Analysis Using Advanced OTFT Biosensor Technology" Doklady Physics Journal(**Published by Doklady Physics, SCI, Peer reviewed, Impact Factor: (0.6) Article id:DANPhys 2660085Tiwari) on 29 June 2026**

INTERNATIONAL CONFERENCE

[7] **Sandeep Kumar Ojha** and Brijesh Kumar, 2020, "Performance Analysis and Parameter Extraction of Organic Light Emitting Transistor (OLET)", **IEEE Proc. Int. Conf. on Electrical and Electronics Engineering (ICE3-2020), Gorakhpur, pp. 711-716**, February 14-15, 2020 (Published by IEEE and Available Online through IEEE Xplore).

[8] **Sandeep Kumar Ojha** and Brijesh Kumar, "Impact analysis on essential parameter of dual gate OTFT" RTU TEQIP-III Sponsored **International Conference on Communication and Intelligent System (ICCIS 2020)**, Jointly organized by Birla Institute of Applied Sciences (BIAS) Uttarakhand and Soft Computing Research Society, Dec. 26-27, 2020.

PERSONAL DOSSIER

Father's Name: Dr. P. N. Ojha

Mother's Name: Smt. Shobha Rani Ojha

Date of Birth: 16th December 1976

PAN No: AALPO2312E

AADHAR No: 4997 4192 5568

Blood Group: AB+



(SANDEEP KUMAR OJHA)

Motivational Letter

Dr. Sandeep Kumar Ojha

Dear Sir/Madam,

I am writing to express my strong motivation to contribute to your esteemed institution as an academician and researcher. My educational journey, research experience, and passion for teaching have shaped my commitment to advancing knowledge in the fields of Microelectronics, VLSI Design, Artificial Intelligence, Internet of Things (IoT), and Semiconductor Technologies.

I hold a Ph.D. in Microelectronics and VLSI Design, an M.Tech. (Honors) in VLSI Design, and a B.Tech. in Electronics and Telecommunication Engineering. Throughout my academic career, I have consistently pursued excellence in teaching, research, and innovation. My doctoral research focused on analytical modeling and parameter extraction of Organic Light-Emitting Transistors (OLETs), providing valuable contributions to emerging semiconductor device technologies.

Teaching has always been my greatest passion. I believe that education is not merely the transfer of knowledge but the inspiration of curiosity, creativity, and critical thinking. Over the years, I have taught undergraduate and postgraduate students in subjects including VLSI Design, Analog and Digital Electronics, IoT, Artificial Intelligence, Cloud Computing, Embedded Systems, Python Programming, and Electronic Devices. I continuously strive to create an engaging learning environment by integrating theoretical concepts with practical applications, industry-oriented projects, and modern digital tools.

My research interests include semiconductor device modeling, nanoelectronics, low-power VLSI systems, intelligent IoT applications, embedded AI, machine learning, smart sensor technologies, and sustainable electronic systems. I aspire to establish interdisciplinary research collaborations that address real-world challenges through innovative engineering solutions. Publishing in high-impact journals, securing competitive research funding, mentoring doctoral scholars, and developing industry partnerships are among my long-term academic objectives.

Persistence, honesty, hard work, and lifelong learning have been the guiding principles of my professional journey. Every challenge has strengthened my determination to contribute meaningfully to higher education and scientific research. I remain committed to creating knowledge that benefits society while inspiring the next generation of engineers and innovators.

I am confident that my academic background, teaching experience, research potential, and dedication to excellence will enable me to make valuable contributions to your institution. I would be honored to become part of your academic community and contribute to its vision of excellence in education, research, and innovation.

Thank you for considering my application. I look forward to the opportunity to contribute to your esteemed institution and grow as a teacher, researcher, and lifelong learner.

Yours sincerely,

Dr. Sandeep Kumar Ojha

ASSISTANT PROFESSOR (vnit, Nagpur)
Ph.D. (Microelectronics & VLSI Design)
M.Tech. (Honors) – VLSI Design
B.Tech. – Electronics & Telecommunication Engineering